Staff answers · retrieval and model

Starter: a gene is a section of DNA coding for a protein [A2 W4]; stem cells divide and some specialise [A2 W2]; chromosomes are in the nucleus [A1 W1]. Confirm actual taught coverage.

Full model and fade: Aa, aa / Aa, aa. Missing second row Aa, aa. aa is 2/4 = 1/2 = 50%. The aa parent can pass only a. Model chance applies per offspring, never an exact family count.

Hinge A is correct. B counts Aa rather than aa. C counts all except aa. D treats possible as certain. Hear both pupils separately. If either is unsure, re-model one top and one side allele for that pupil while the other transfers.

Parallel check: AA × aa gives Aa in all four boxes. Aa has a 100% chance under this model. This does not set a family size.

Supported completes grid; Independent counts and justifies; Aspire rejects guaranteed two aa among four real offspring. Go deeper: Aa × Aa gives 1/4 aa, Aa × aa gives 1/2 aa. Faulty A,A headings wrongly remove the aa chance; restore A,a.

Both exit responses must be individual. Check grid, probability, limit, actual feedback and chosen next step. No grade or evidence code follows automatically from a click.

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Staff · source and teaching care

Weekly binding: Complete a genetic cross (digital Punnett square).

Exact pupil goal: I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

Source: Launch/Autumn_2_2026-27/SCI_LAUNCH_A2_W05/SCI_LAUNCH_A2_W05.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d.

Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B43 Aut2·W5; C43 Complete a genetic cross (digital Punnett square).

Sources are inherited from the verified lesson. No new specification alignment is asserted. [AWARD] [QUAL] names remain masked. No biological or personal family data is used.

Hearing both pupils matters. A correct click is not evidence of reasoning. Re-model the uncertain step for one pupil while the other attempts a fresh example. Preserve the original answer and the prompt used.

A genetic counsellor explains chance without promising a fixed outcome. This is a career link, not advice about a person.

Model care: This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

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